

Skill dependencies underlie career paths and have societal implications

A framework to understand the labour market reveals a nested hierarchical architecture in human capital in which specific knowledge and skills are contingent upon foundational, general skills and knowledge. This nested skill structure provides a new perspective on wage premiums and persistent wage disparity observed across different demographic groups.

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The problem

The modern economy demands a diverse array of complex skills, knowledge, and abilities that encompass critical thinking, social capabilities and technical expertise. Some of these traits – especially those that are valuable and well-compensated – are highly specialized and difficult to acquire, and they often require extensive educational and career development¹.

This specialization is not cumulative and sequential but follows a nested dependent structure. For example, take physics, in which specializing in Einstein's theory of relativity requires an understanding of calculus, which itself relies on knowledge of algebra and geometry – all of these requirements are underpinned by basic logical thinking skills. As seen in educational curricula, advanced skills and knowledge are often contained within or built upon broader, fundamental skills.

This nested hierarchical nature of skill development (and thus of human capital) raises questions about its effect on the labour market; for example, how this complex web of interdependent skills manifests, and who benefits from or might be left behind by this structure.

The discovery

We infer the skill dependencies of different jobs from the results of US workplace and household surveys along with 70 million job transition records from resumes, which reveals a nested hierarchical structure in human capital (Fig. 1a). This discovery offers two key insights. First, skills are acquired and applied through interconnected learning stages rather than in isolation, which shapes skill development pathways and career progression. Second, broadly applicable general skills constitute the foundation across occupations, upon which more-specific skills are built later in career paths. These findings are consistent with our common understanding of how higher-value jobs become accessible through prerequisite skills acquired or applied along these nested paths.

Although the nested hierarchy itself is not surprising, its systemic implications are notable. We quantified the alignment of each skill within the overall nested structure, and we calculated how well it fit into dependency chains. We find that skills that are more closely aligned with the nested hierarchy require a longer education, command higher wage premiums and are less likely to be automated

(Fig. 1b, c). This observation is true even when managerial roles or social skills (which are broadly associated with job seniority) are not included in the analysis, which highlights the fundamental nature of the mechanism. Our findings suggest that the value and properties of a skill are largely determined by its position within the overall skills structure, which potentially further reinforces the nested hierarchy. Indeed, this nested structure has become even more pronounced over the past two decades, which suggests that there might have been an increase in polarization within the job landscape.

The implications

Our nested dependency structure introduces a new dimension to human capital and moves beyond traditional one-dimensional metrics such as years of schooling or work experience. Additionally, the directional nature of these nested dependencies suggests that the development of human capital follows structured pathways. The nested dependency structure itself is not entirely surprising, as it is evident in curricula and job-training manuals. However, understanding skills through our structural alignment departs from traditional context-informed analysis in social science, and thereby it complements existing frameworks in human capital theory. Thus, this study offers insight into how contextual factors can be defined as structural factors.

In addition to these theoretical insights, our context-free nested hierarchical structure enables us to easily track the structural shifts in workplace skills, capabilities and tools. In today's constantly evolving labour market, in which new skills emerge and older skills become obsolete, it is useful to track how shifts in the structure affect long-term wage penalties across different demographics.

As society becomes increasingly complex and interdependent, the dependency structures of required knowledge and skills become more nested, which creates greater distinctions between those who can access these layers and those who cannot. As this hierarchical structure fundamentally shapes our social and economic systems, systematically understanding these nested structures is crucial for developing timely and effective policies that ensure equitable access to opportunities for developing fundamental skills across all demographic groups and regions.

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EXPERT OPINION

"I am very excited by this paper. I believe that it makes a unique and important theoretical contribution to our understanding of human capital development. The authors develop a novel

theory, which they explore using relevant and informative empirics. The study is exceptionally thorough." **Christopher Esposito, UCLA, Los Angeles, CA, USA.**

FIGURE

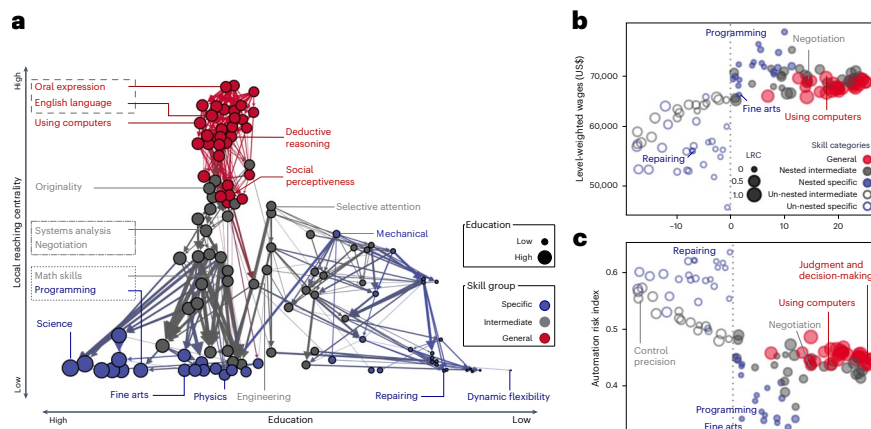


Fig. 1 | Skill dependency hierarchy. a, Skills (nodes) are connected through the occupations they are required for; the direction of the arrow (from skill A to skill B) indicates that the application or acquisition of skill B is more likely to be contingent on skill A than vice versa. Each skill is positioned based on educational attainment (x axis) and local reaching centrality (y axis), defined as the proportion of skills reachable from it, indicating how many specific skills build upon the skill. b, The structural alignment of skills within the nested structure correlates with their expected wages. c, The structural alignment of skills within the nested structure correlates with their risk of automation³. Marker sizes are proportional to the skill's local reaching centrality (LRC) in the hierarchy network. © 2025, Hosseinioun, M. et al., CC-BY 4.0.

BEHIND THE PAPER

Our research began in 2020 with a simple question: how can combined skills create more value than isolated ones? However, we soon discovered an additional dimension beyond traditional economic mechanisms of complementarity^{3,4}. Skills are acquired and applied together not just because they are complementary, but because they are interdependent, indicating progressions that explain why certain career choices lead to irreversible transitions or unlock new opportunities. Excited by this discovery, we sought to develop it further, but hit a roadblock in quantifying this nested structure.

A breakthrough came from an unexpected source. A paper by colleagues in systems ecology, studying interactions between generalists and specialists, provided the key⁵. We adapted their methods (originally used to analyse species on islands) to occupational skills, and the results were illuminating. We found the presence of a nested structure in human capital just as in ecosystems; this structure rewards skills differentially, leading to substantial disparities and long-term wage penalties across demographic groups. **H.Y.**

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FROM THE EDITOR

"When this work was submitted to us, it struck us for both its novel conceptualization of human capital and how occupational skills develop over time, as well as the scale of empirical evidence used to support the theory." **Editorial Team, Nature Human Behaviour.**